

Overview on Information Security

Ngô Minh Nhựt

2024

Outline

❑ **Cryptography**

- Public key encryption
- Digital signature
- TLS and HTTPs

❑ Cyber threats

❑ Blockchain

- Cryptocurrency
- Decentralized application
- Web 3.0

Cryptography

- ❑ Cryptography is the practice and study of techniques for secure communication by converting plain text into ciphertext
 - Prevent third parties or the public from reading private messages
- ❑ Heavily based on mathematics and computer science
 - Algorithms are designed around computational hardness, making them hard to break

Roles of cryptography

- ❑ Confidentiality: ensures that information is accessible only to authorized individuals or systems
 - Encryption: converts plaintext into ciphertext, making it unreadable to unauthorized users
- ❑ Authentication: approve or deny users to access a system or a resource
 - Common form of authentication is using passwords
 - Hashing password: passwords are hashed before storing

Roles of cryptography

- Integrity: ensures that data have not been altered or tampered during transmission or storage
 - Hash functions: generate a fixed-size hash value (digest) from input data
 - Digital signatures: use hashing combined with asymmetric encryption to provide a verifiable proof of data integrity and authenticity

Roles of cryptography

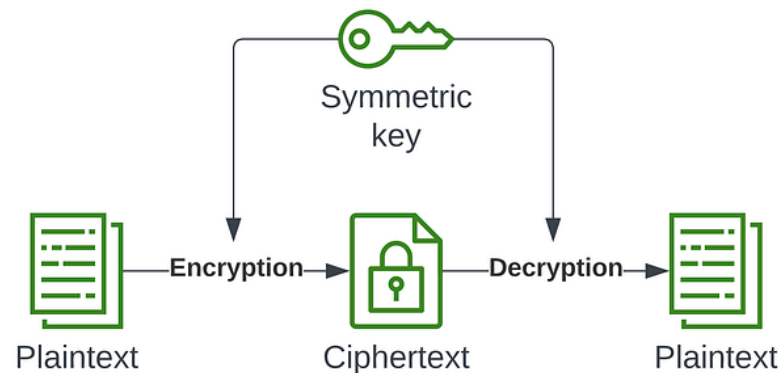
- ❑ Secure communication: protects data transmitted over networks from eavesdropping and tampering
 - TLS/SSL: protocols that use cryptographic techniques to secure data transmitted between clients and servers
 - Protects the integrity and confidentiality of financial and sensitive transactions

Roles of cryptography

- ❑ Data protection: protect sensitive data from unauthorized access in case of physical theft
 - File and disk encryption: encrypt files or entire disks
 - Database encryption: encrypt database fields or entire databases

Symmetric key encryption

- Data are encrypted and decrypted using the same key



- Low computational complexity
- Key exchange problem

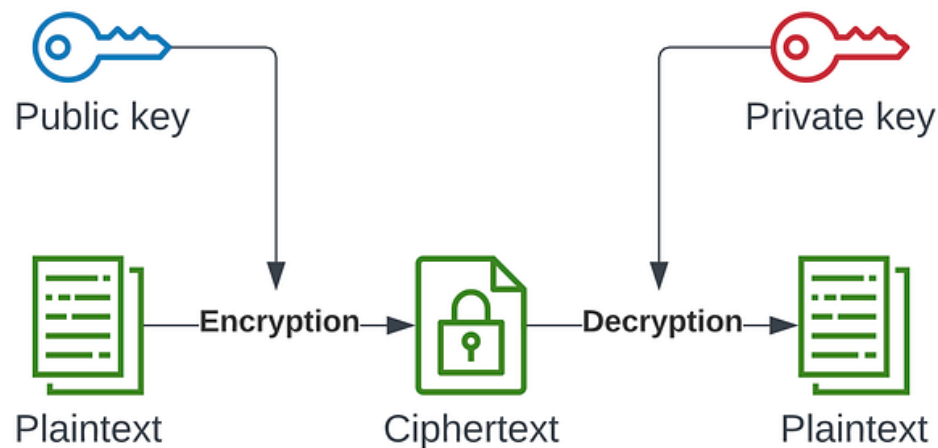
Source: <https://medium.com/@igorfilatov>

Key exchange problem

- ❑ How can receiver get the key?
 - Go to the sender's house to get the key?
 - Can we encrypt with **key A** and decrypt with **key B**?
- ❑ Public key vs private key
 - Public key is shared
 - Private key is kept secret

Asymmetric key encryption

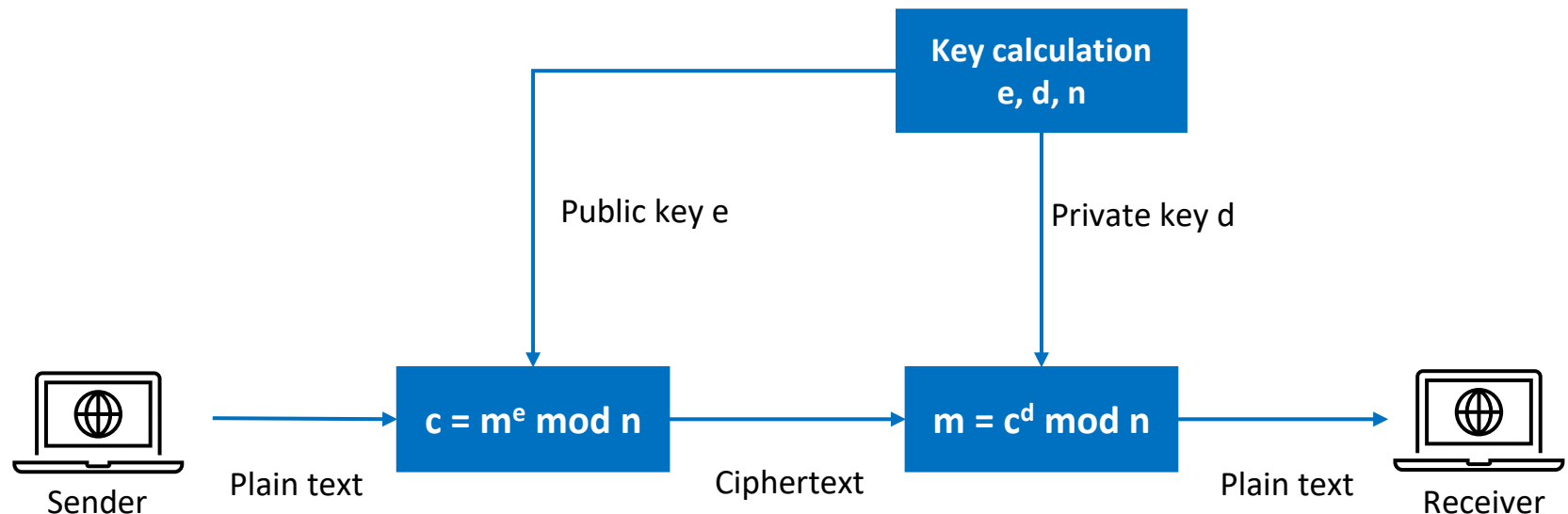
- ❑ Message is encrypted using the recipient's public key.
- ❑ Only the recipient's private key can decrypt the message.



Source: <https://medium.com/@igorfilatov>

Asymmetric key encryption

- ❑ Public key vs private key
 - These keys are different but mathematically related
- ❑ RSA algorithm



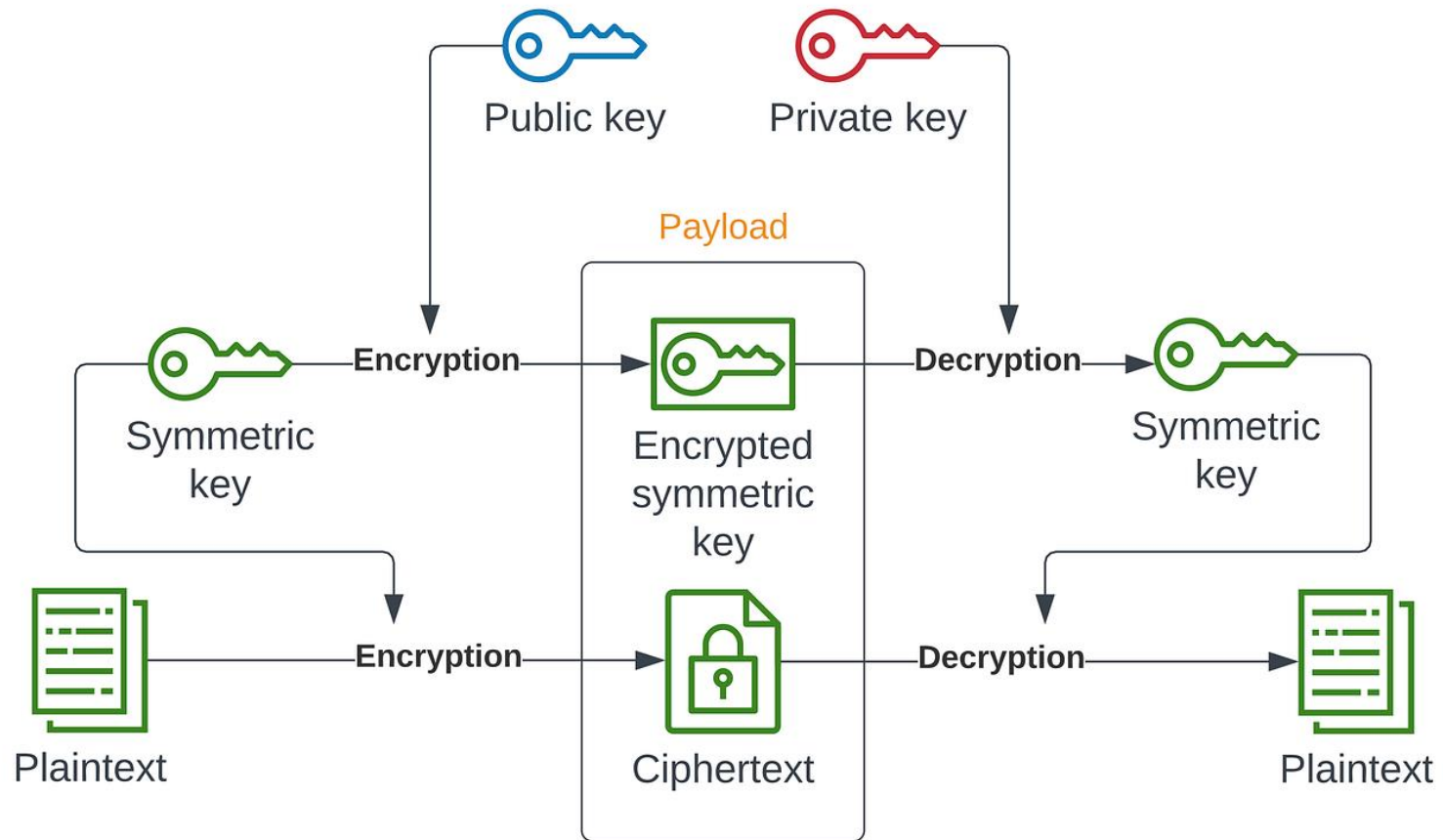
Asymmetric key encryption

- ❑ Challenges and limitations
 - Computational intensity
 - Slower than symmetric encryption

Hybrid encryption

- ❑ Combine public key cryptography with symmetric encryption (like AES) for efficiency
- ❑ Public key encryption secures the symmetric key, while AES encrypts the actual data

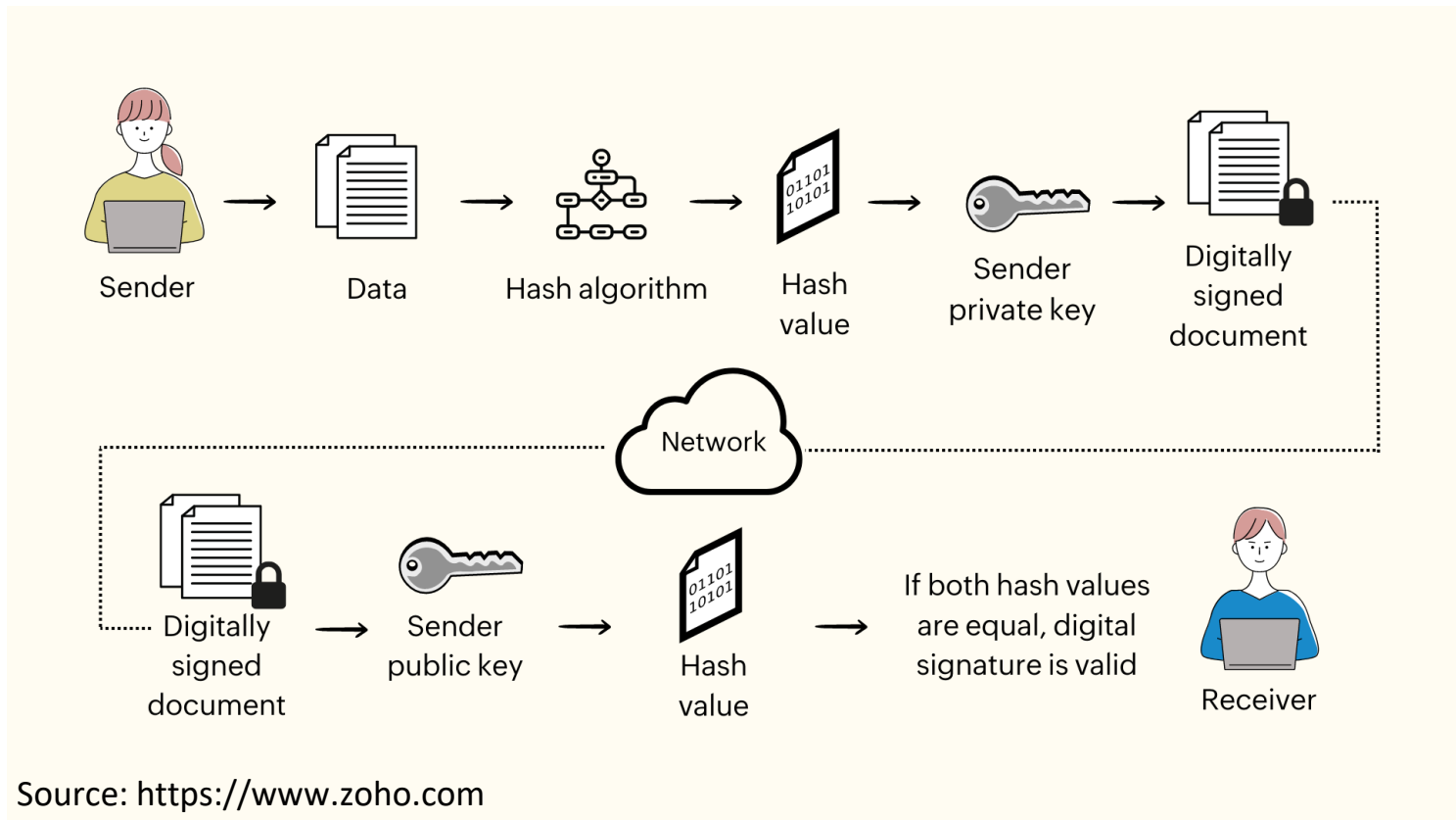
Hybrid encryption



Source: <https://medium.com/@igorfilatov>

Digital signature

- ❑ Private key is used to sign
- ❑ Public key is used to verify



Hypertext Transfer Protocol Secure

- ❑ HTTP: unencrypted data can lead to sensitive information like passwords or credit card details stolen
- ❑ HTTPS: secure web browsing with encryption
- ❑ How do web server and client exchange key?

Hypertext Transfer Protocol Secure

- ❑ Transport Layer Security (TLS)
 - Define protocol for data encryption, e.g., encryption method, server certificate, encryption key, etc.
- ❑ Certificate authority (CA)
 - Certify web server so that web browser know it is communicated with a trusted/right one

Public key encryption

□ Key takeaways

- Public and private key roles
- Hybrid encryption for efficiency
- Importance in modern-day security

□ Future challenges

- Security is ensured by the difficulty of key searching
- Quantum computing with its power can break public key cryptography

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❑ **Cyber threats**

❑ Blockchain

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- Decentralized application
- Web 3.0

Cyber Threats

- ❑ Malware: malicious software designed to harm or exploit systems, such as viruses, worms, and ransomware
- ❑ Zero-day vulnerabilities: vulnerabilities that there is rules/patterns not exposed
- ❑ Ransomware

Cyber Threats

- ❑ Phishing: attempts to obtain sensitive information by disguising as a trustworthy entity, often through email or social engineering
- ❑ Insider threats: security risks posed by individuals within an organization
 - Such as employees or contractors, who may misuse their access to information

Cyber Threats Evolution

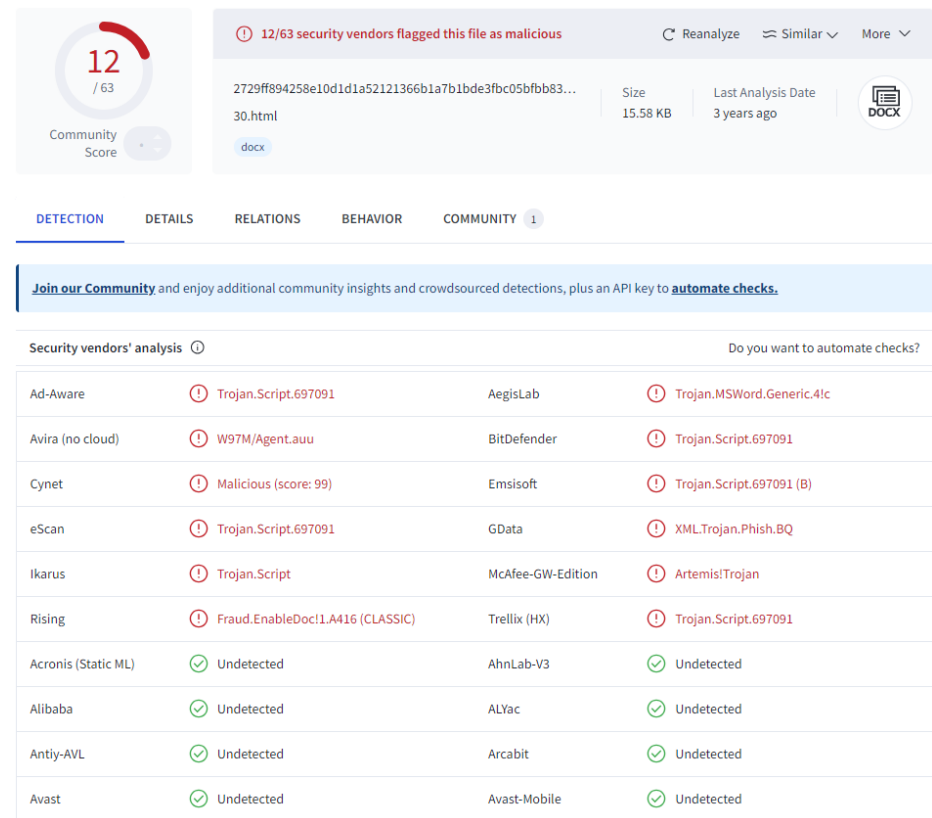
- ❑ Increasing sophistication of cyber attacks, such as advanced persistent threats (APTs), zero-day vulnerabilities, ransomware, phishing, social engineering
- ❑ Attack vectors
 - ❑ Vulnerable applications
 - ❑ Web applications or services
 - ❑ Files or documents with active content, e.g., Word document

Malware detection

- ❑ Rule-based detection
- ❑ Detection with machine learning

Malware detection efficacy

- ❑ Detection rate is low
- ❑ Take time to respond to new malware
- ❑ Multi-detection: scan files with multiple AV software



Source: <https://www.virustotal.com>

Content disarm and reconstruction

- ❑ CDR works by break files into components, process them and reconstruct processed ones
- ❑ Prevention is better than detection
 - Potentially malicious components are removed or neutralized
- ❑ Effective against zero-day attacks

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Blockchain

- ❑ Blockchain is a shared and immutable ledger that facilitates the process of recording transactions and tracking assets
 - List of records/blocks that are securely linked together via cryptographic hashes
 - **Each block** contains a **cryptographic hash** of the previous block, a **timestamp**, and **transaction data**

Blockchain

□ Key features of blockchain

- Immutability: data cannot be altered once it is recorded
- Transparency: all transactions are visible to participants
- Security: secure transactions are ensured by cryptography
- Smart contract: self-executing piece of code (contract) stored on blockchain
- Decentralization: no single authority controls the network

Cryptocurrency

- ❑ A form of currency that exists digitally or virtually and uses cryptography to secure transactions
- ❑ Do not have a central issuing or regulating authority. Instead use a decentralized system to record transactions and issue new units

Cryptocurrency

- ❑ Cryptocurrencies are generally viewed as a distinct asset class
 - Not considered to be currencies in the traditional sense
- ❑ The validity of each cryptocurrency's coins is provided by a blockchain
 - Proof-of-work: transactions are validated by distrustful parties referred to as miners
 - Proof-of-stake: transactions are validated by holders of the associated cryptocurrency, sometimes grouped together in stake pools

Decentralized application

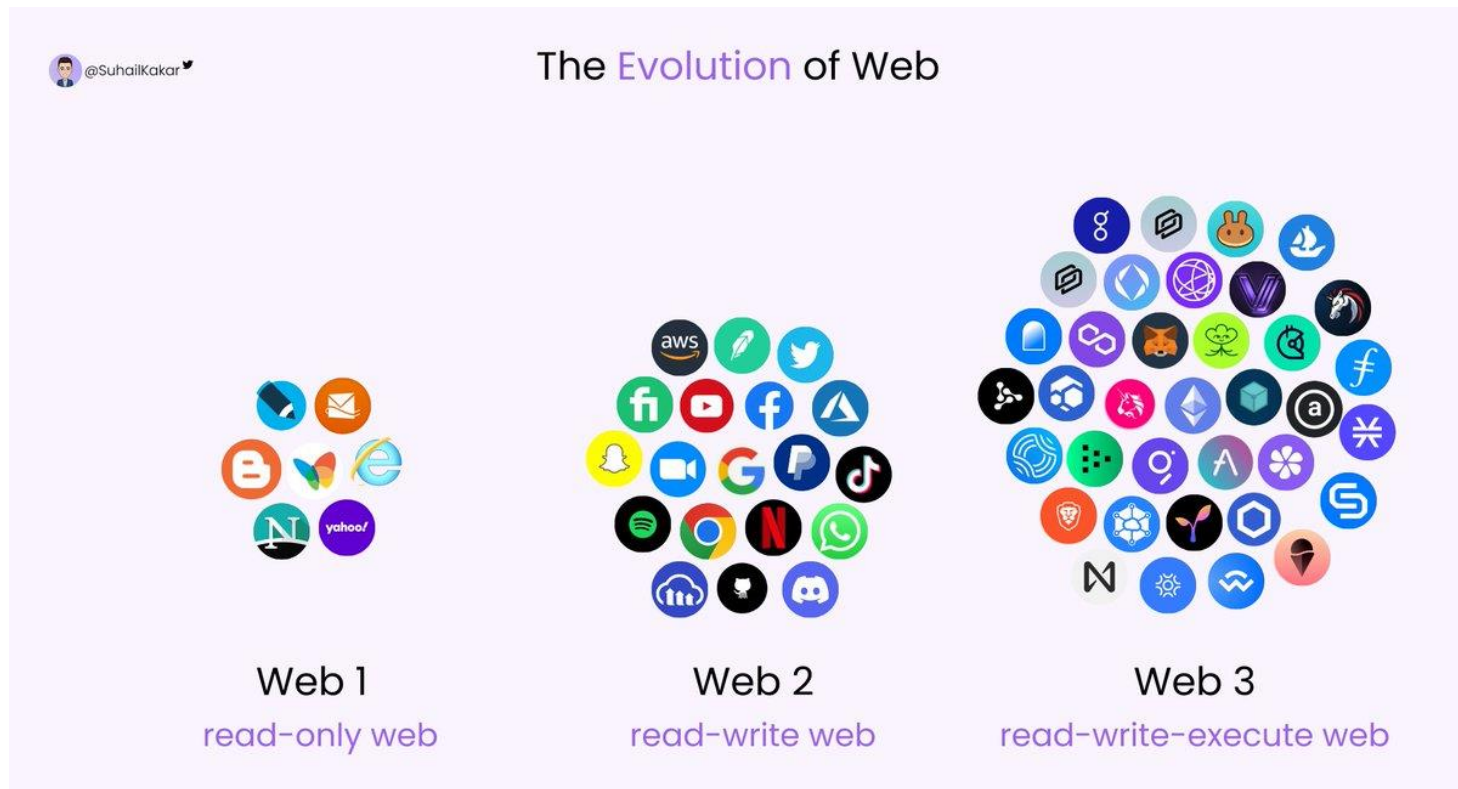
- ❑ Decentralized application: DApp
- ❑ Software programs that run on a blockchain network of computers instead of on a single computer
- ❑ Unlike traditional applications, DApps operate without human intervention and are not owned by any one entity

Smart contract

- ❑ Self-executing piece of code stored on a blockchain, where the terms and conditions of the contract are directly written into code
- ❑ Smart contract code is visible and verifiable on the blockchain. It is trusted that the contract will execute as agreed without needing a trusted third party

Web 3.0

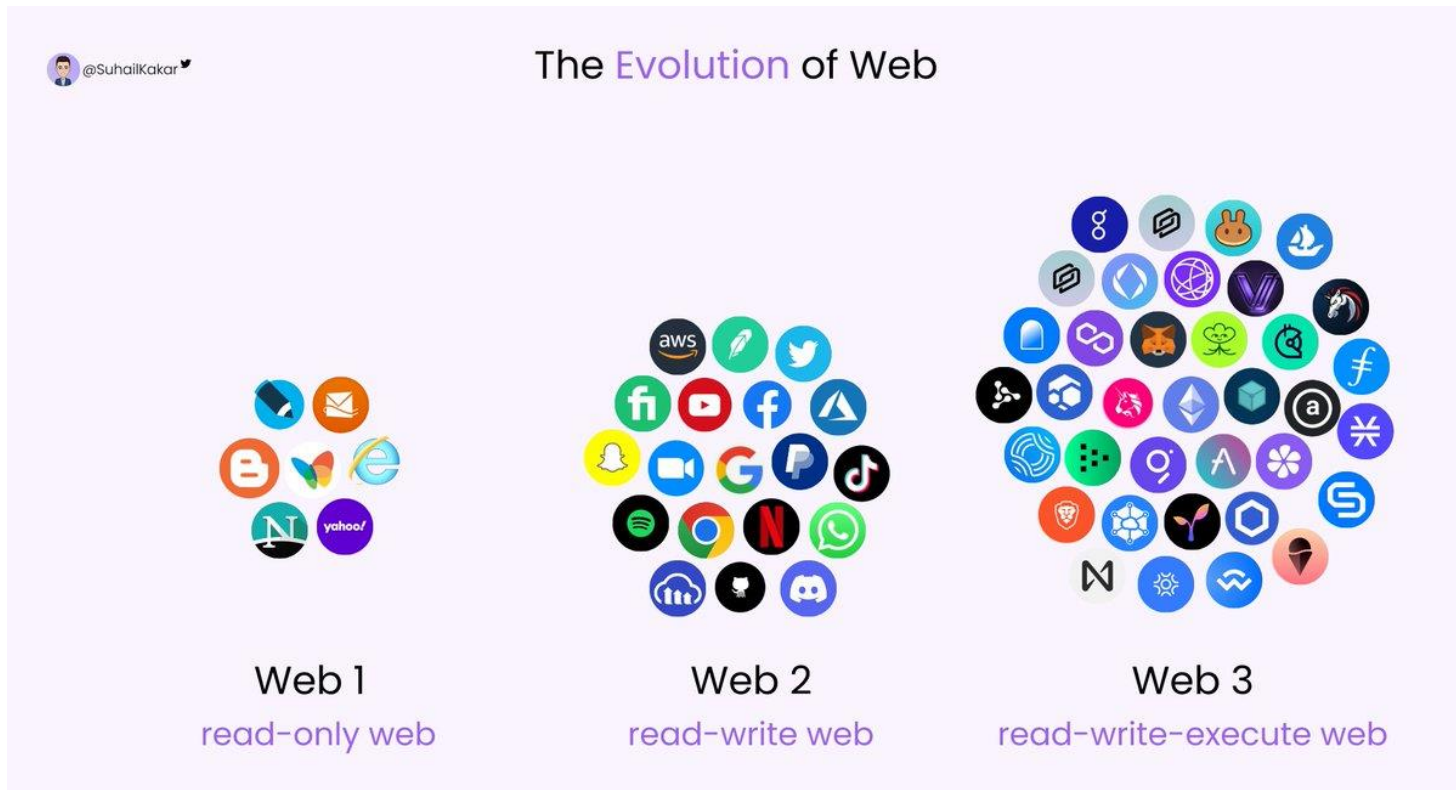
- ❑ Web 2.0: problem of privacy, confidentiality and decentralization



Source: Internet

Web 3.0

- ❑ Web 3.0: operate on blockchain, bring ownership and authority to users



Source: Internet

Challenges and limitations

- ❑ Each transaction requires a huge amount of computational power, leading to two major issues
 - Scalability: limited transaction speed
 - Energy consumption: proof-of-work specially consumes energy

Research topics in information security

- ❑ Security of Wireless Sensor Networks
- ❑ Elliptic-curve cryptography (ECC)
- ❑ Blockchain and applications with privacy preservation
 - Decentralized applications and Web 3.0
- ❑ Machine learning models for malware detection
- ❑ Data mining with privacy preservation
- ❑ Privacy-preserving federated learning with differential privacy